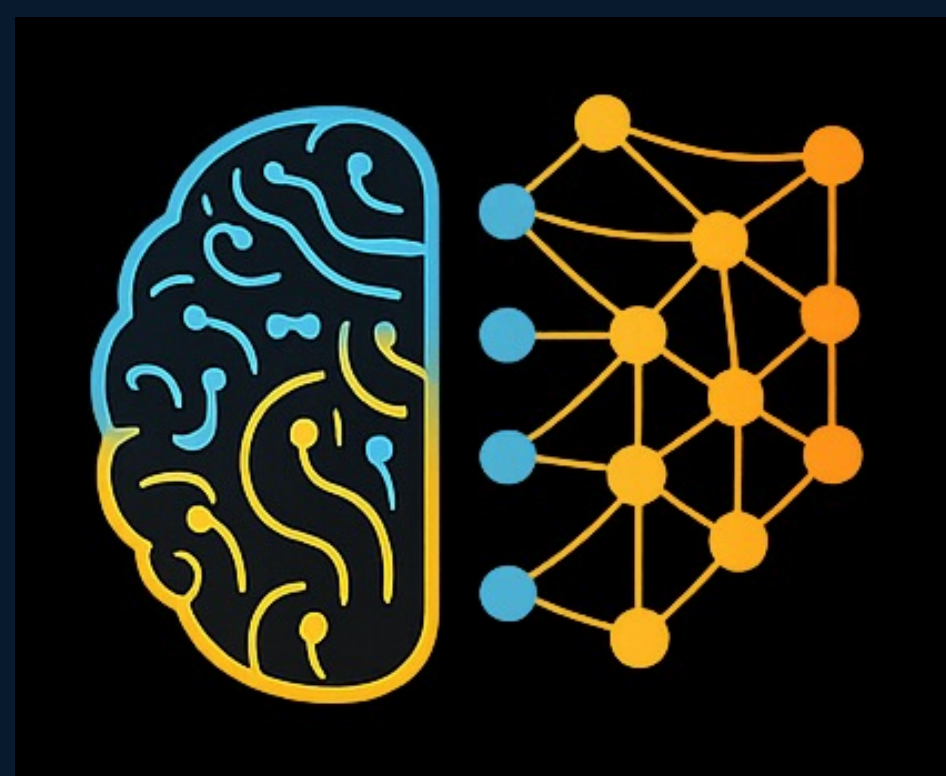




GrIDDD

Graph Diffusion that can Insert and Delete

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1. Introduction

From fixed-size noise to adaptive molecular graphs.

Discrete graph diffusion models (DDPMs) are among the most principled approaches to molecular generation. However, they share a fundamental limitation: **the graph size is fixed** before generation starts and never changes during the process. This means the model cannot add or remove atoms at any point; the final molecule will always have exactly as many atoms as the initial noise sample.

Why we care?

This becomes critical in two scenarios:

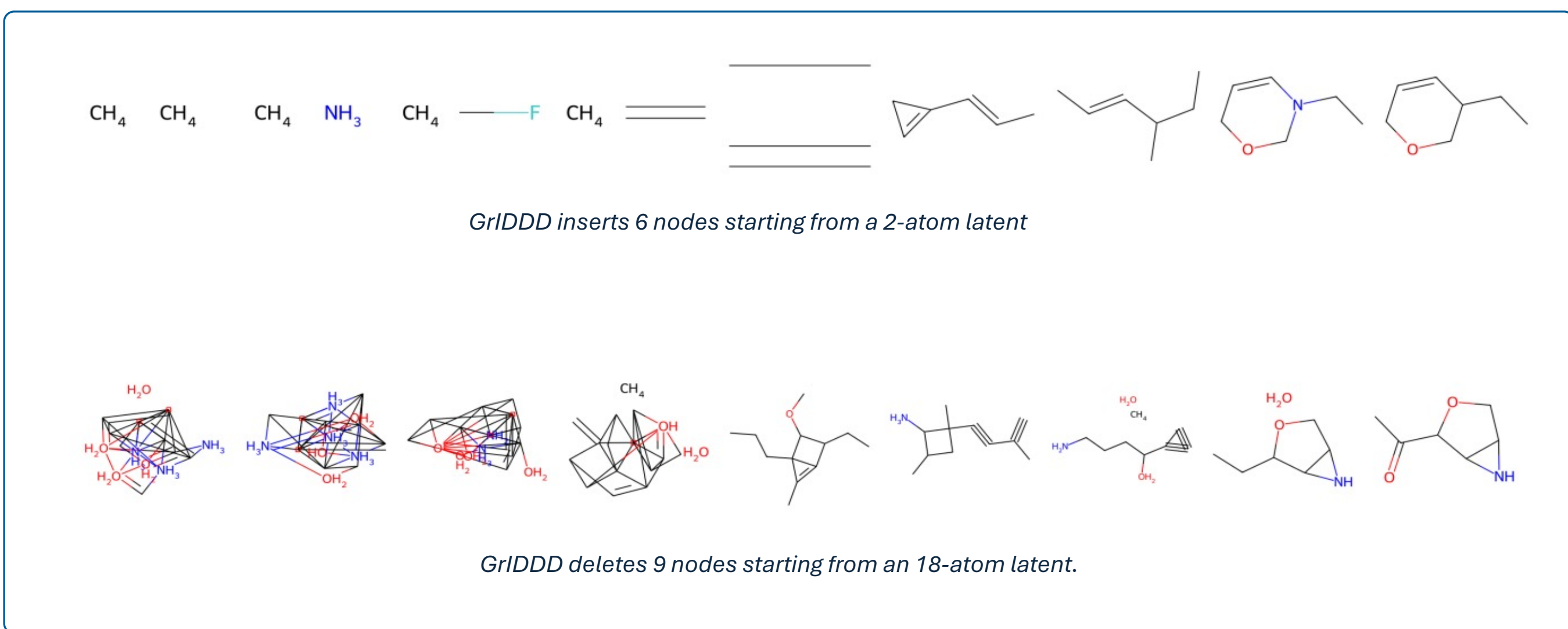
- Property targeting:** properties like molecular weight or LogP correlate directly with atom count, a fixed size prevents the model from adapting its structure to the target, unless the correct size is somehow known in advance. Existing models like DiGress and FreeGress circumvent this by sampling the graph size before generation, but this requires knowing the correct size in advance.
- Property optimization:** improving a molecule often requires adding or removing atoms entirely, which is impossible when graph size is fixed by design. Existing approaches simply cannot address this scenario.

Contribution

GrIDDD (Graph Insert-Delete Discrete Diffusion) is the first graph diffusion model that **dynamically changes the graph size during denoising**, through monotonic node insertions and deletions.

Unlike previous models that sample the graph size before generation begins, GrIDDD learns to adapt it dynamically as part of the denoising process itself.

GrIDDD matches or exceeds state-of-the-art performance on property targeting, and significantly outperforms existing methods on property optimization, despite being trained on a harder problem.



2. Approach

GrIDDD, generalized to monotonic node insertions and deletions to support dynamic graph resizing.

Data Pipeline: Molecules are represented as one-hot encoded tensors $\mathbf{G} = (\mathbf{X}, \mathbf{E})$, where $\mathbf{X}$ denotes node (atom) types and $\mathbf{E}$ encodes bond connectivity/types. The generative process is conditioned on target property $\mathbf{y}$ via classifier-free guidance.

1. Primary Objective & Dynamic Resizing :

The objective of the GrIDDD framework is to generate optimized molecular graphs by supporting dynamic graph resizing during diffusion through two core operations:

- Monotonic Deletion:** Forcefully transitioning valid atoms into a permanent empty state.
- Monotonic Insertion:** Activating nodes from an empty latent space to remain permanently valid.

2. The Forward Process :

To adjust the graph, the difference ($\Delta\mathbf{T}$) between the target noisy graph size $\mathbf{n}^1$ and the clean graph size $\mathbf{n}^0$ is calculated.

- $\Delta\mathbf{T} = \mathbf{0}$ Standard diffusion, no structural changes.

- $\Delta\mathbf{T} < \mathbf{0}$ Atoms transition into absorbing DEL state via DEL* intermediate.

$$\mathbf{Q}^{\mathbf{t}} = \zeta(\mathbf{t})(\mathbf{A}^{\mathbf{t}}\mathbf{A}^* + (\mathbf{1} - \mathbf{A}^{\mathbf{t}})\mathbf{B}^*) + (\mathbf{1} - \zeta(\mathbf{t}))\mathbf{C}^*$$

- $\Delta\mathbf{T} > \mathbf{0}$ New nodes activated and initialized from marginal distribution $\mathbf{m}_\mathbf{X}$ (for node's state) and $\mathbf{m}_\mathbf{E}$ (for node's edges).

The timing for inserting or deleting nodes is scheduled using a logistic distribution's probability density function $\zeta'(\mathbf{t})$, while $\mathbf{n}^T$ is sampled from a discrete distribution on the number of nodes $\mathbf{h}_{\mathbf{n}^0}(\mathbf{n})$ to determine the target size of the fully noisy graph where $\mathbf{n}_0$ (original graph size)

$$\zeta'(\mathbf{t}) = \left| \frac{\frac{\mathbf{t}-\mathbf{D}}{\mathbf{e}^{\mathbf{w}}} }{(\frac{\mathbf{t}-\mathbf{D}}{\mathbf{e}^{\mathbf{w}}} + 1)^2} \right| \quad \mathbf{h}_{\mathbf{n}^0}(\mathbf{n}) = \mathbf{p}_{\max} + \frac{\mathbf{p}_{\min} - \mathbf{p}_{\max}}{\mathbf{n}_{\max}} |\mathbf{n} - \mathbf{n}^0|$$

3. The Reverse Process:

The model predicts an *activation time* $\mathbf{s}$ per node via a generalized posterior, triggering exact insertions and deletions to reconstruct the clean graph.

$$\mathbf{p}_\theta(\mathbf{x}^{\mathbf{t}-1} | \mathbf{x}^{\mathbf{t}}, \mathbf{y}) = \sum_{\mathbf{x}} \frac{q(\mathbf{x}^{\mathbf{t}}|\mathbf{x}^{\mathbf{t}-1})q(\mathbf{x}^{\mathbf{t}-1}|\mathbf{x}^{\mathbf{s}} = \mathbf{x})}{q(\mathbf{x}^{\mathbf{t}}|\mathbf{x}^{\mathbf{s}} = \mathbf{x})} \mathbf{p}_\theta(\mathbf{x}^{\mathbf{s}} = \mathbf{x} | \mathbf{x}^{\mathbf{t}}, \mathbf{y})$$

4. Architecture, Training, and Sampling

Training Strategy: A primary network ($\mathbf{p}_\theta$) is optimized via Cross-Entropy loss to reconstruct the original node/edge categories and to predict the activation time ($\mathbf{s}$) for each node. Concurrently, an auxiliary network ($\mathbf{g}_\theta$) is trained to predict the exact number of transient **DEL*** states required at each timestep to facilitate future insertions.

Sampling Generation: At generation time, sampling initializes from a fully noisy latent graph $\mathbf{G}^T$ of size $\mathbf{n}^T$. As the generalized posterior iteratively denoises the structure, changes in graph size are executed dynamically. To reverse forward deletions, $\mathbf{g}_\theta$ appends **DEL*** placeholders that the main network populates with valid atom types. Conversely, to reverse forward insertions, if the model predicts that a node's activation time matches the current timestep ($\mathbf{s} = \mathbf{t}$), that node is permanently excised from the graph. By $\mathbf{t} = \mathbf{0}$, this balanced process yields a clean, chemically valid, and size-optimized molecule tailored to the conditioning property $\mathbf{y}$.

Transition matrices used to compute $\mathbf{Q}^{\mathbf{t}}$ and $\overline{\mathbf{Q}}^{\mathbf{t}|\mathbf{s}}$. For formatting purposes, the DEL and DEL* states are abbreviated as D and D*.

$$\mathbf{A}^* = \begin{bmatrix} \mathbf{A} & \mathbf{D} & \mathbf{D}^* \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \mathbf{B}^* = \begin{bmatrix} \mathbf{B} & \mathbf{D} & \mathbf{D}^* \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \mathbf{C}^* = \begin{bmatrix} \mathbf{C} & \mathbf{D} & \mathbf{D}^* \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad \mathbf{D}^* = \begin{bmatrix} \mathbf{D} & \mathbf{D}^* \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

- $\mathbf{A}^*$ (**Identity**): Nodes stay exactly as they are.
- $\mathbf{B}^*$ (**Marginal**): Applies normal diffusion noise (randomly swaps valid atom types).
- $\mathbf{C}^*$ (**Deletion**): Actively forces valid atoms into the empty (DEL) state.
- $\mathbf{D}^*$ (**Total Wipeout**): An extreme state forcing all nodes immediately into the DEL state.

$$\overline{\mathbf{Q}}^{\mathbf{t}|\mathbf{s}} = \zeta^{\mathbf{t}|\mathbf{s}} \left(\overline{\alpha}^{\mathbf{t}|\mathbf{s}} \mathbf{A}^* + \left(1 - \overline{\alpha}^{\mathbf{t}|\mathbf{s}} \right) \mathbf{B}^* \right) + \zeta^{\mathbf{t}-1|\mathbf{s}} (1 - \zeta(\mathbf{t})) \mathbf{C}^* + \left(1 - \zeta^{\mathbf{t}-1|\mathbf{s}} \right) \mathbf{D}^*$$

3. Experiments

Datasets, baselines and key results.

Setup

GrIDDD is evaluated on *QM9* (organic molecules of up to nine atoms) and *ZINC-250k* (larger drug-like compounds). Diffusion runs for 500 steps with classifier-free guidance, using $\lambda = 3$ on QM9 and $\lambda = 2$ on ZINC. Asymmetric noise ($\nu = 1$ for nodes, $\nu = 1.5$ for edges) and loss weights $\lambda_{\mathbf{X}} = 1, \lambda_{\mathbf{E}} = 2$; insertion/deletion probability ramps from 0.2 to 1 along the schedule. Performance is measured on two tasks:

targeting, in which the model receives 100 property values and generates 10 molecules per value with MAE as the error metric; and **optimization**, in which (following Jin et al.) an input is corrupted for 100 steps, then denoised toward the target, and the best candidate above a Tanimoto similarity threshold is kept.

Baselines

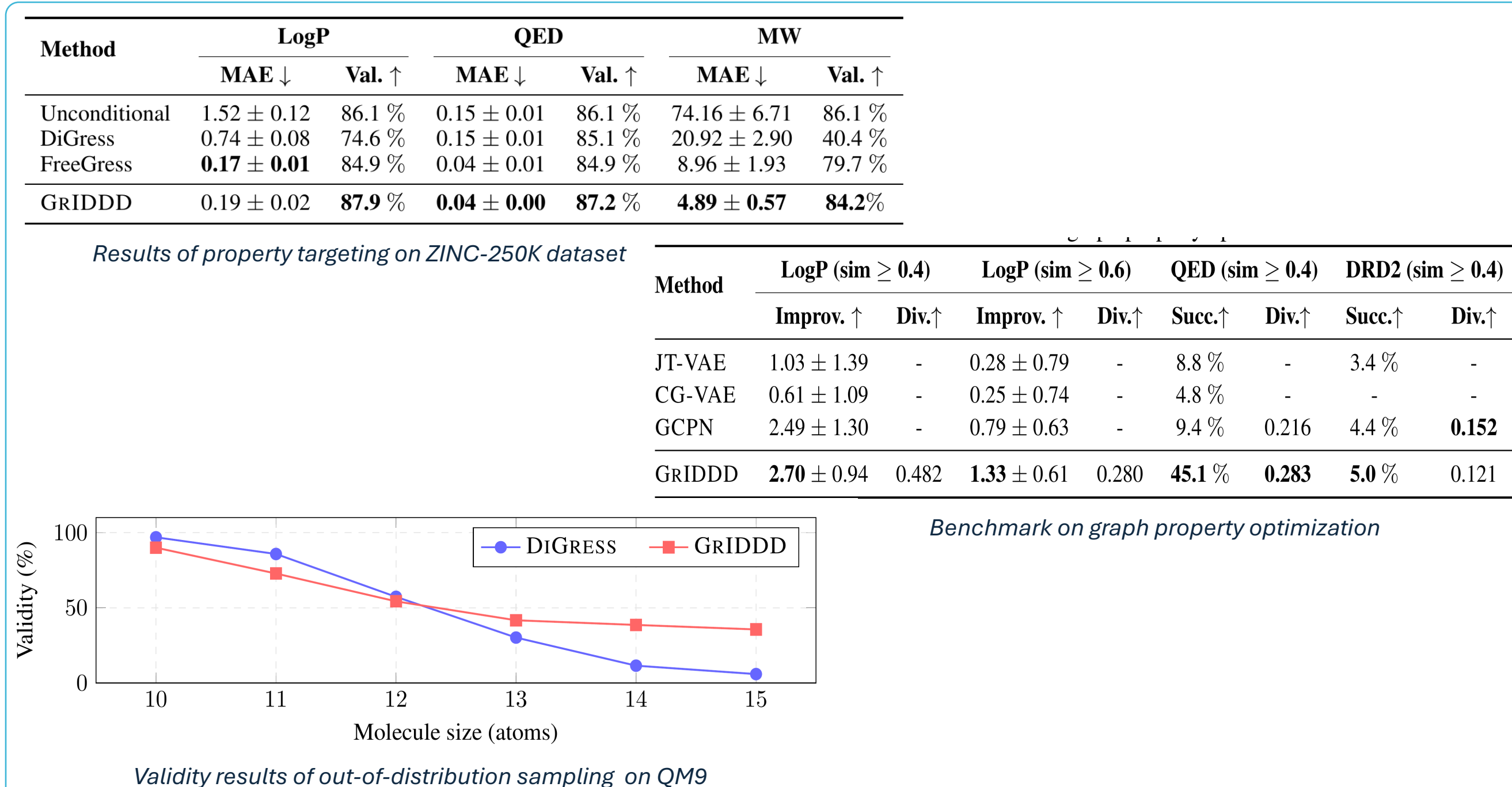
FreeGress and DiGress for targeting; JT-VAE, CG-VAE, GCPN for optimization. All operate on a fixed graph size. GrIDDD is the only model that can insert and delete atoms while sampling.

Targeting accuracy

GrIDDD lowers MAE by about **14% on average**. On the QM9 dipole moment μ it reaches 0.66 against 0.74 for FreeGress, and the gap widens on ZINC, where the molecular weight error falls from 8.96 to 4.89, about half of the strongest baseline's error. Validity stays comparable to or above FreeGress in most settings.

Optimization, OOD, ablation

On constrained optimization with Tanimoto ≥ 0.4 , **QED** success reaches 45.1% against 9.4% for GCPN (target $\in [0.9, 1.0]$); **LogP** improvement is 2.70 at $\delta = 0.6$; **DRD2** lifts low-activity molecules (≤ 0.05) past 0.5. Out of distribution, trained only up to 14 atoms, GrIDDD still produces **35% valid molecules at size 15**; DiGress collapses to near zero. Disabling insertion and deletion drops QED success from 45.1% to 33.8%, confirming that the gain comes from size control rather than from the diffusion backbone. Cost: $\sim 30\%$ slower than FreeGress at training, offset by a tunable number of sampling steps.



4. Personal considerations

Limitations, Assumptions, Lessons learned and Future Work

Limitations

- The two networks composing GrIDDD can **occasionally conflict**, performing an insertion and a deletion at the same timestep, which is theoretically .
- The model also tends to produce more **disconnected** (split) **molecules** than FreeGress or DiGress, likely because atoms inserted late in the denoising process have too few steps left to attach to the main graph.
- Training is **$\sim 30\%$ slower than FreeGress** due to the extra insert/delete bookkeeping, although sampling can actually be faster when starting from small latents. The reduced speed during training is due to the additional complexity of the insert/delete operations.

Assumptions

The model assumes independence between nodes and edges during the diffusion process, as well as independence between nodes themselves in the reverse process. Both are standard approximations in graph diffusion models, but do **not fully reflect** the complexity of **real chemical** bonding.

Lessons learned

The main finding is that the noise scheduler is itself a design lever, not just a hyperparameter to tune: a badly placed peak in $\zeta'(\mathbf{t})$ directly degrades the validity of generated outputs. Getting the shape of $\zeta'(\mathbf{t})$ right matters as much as the architecture choices.

Future work

Future work will explore whether the node **insertion/deletions distribution** can be **learned** end-to-end, replacing the current fixed scheduler ζ . Additionally, since disconnected molecules are likely caused by nodes inserted too late in the denoising process, when too few steps remain to connect them to the main graph, a direction of work is to **provide** the model with **explicit information** about the **current connectivity** of the graph, such as the number of connected components, so it can make better insertion decisions.

